

PROJECT BRIEF

OVERVIEW

Build a webpage that displays real-time date and time and updates every second.

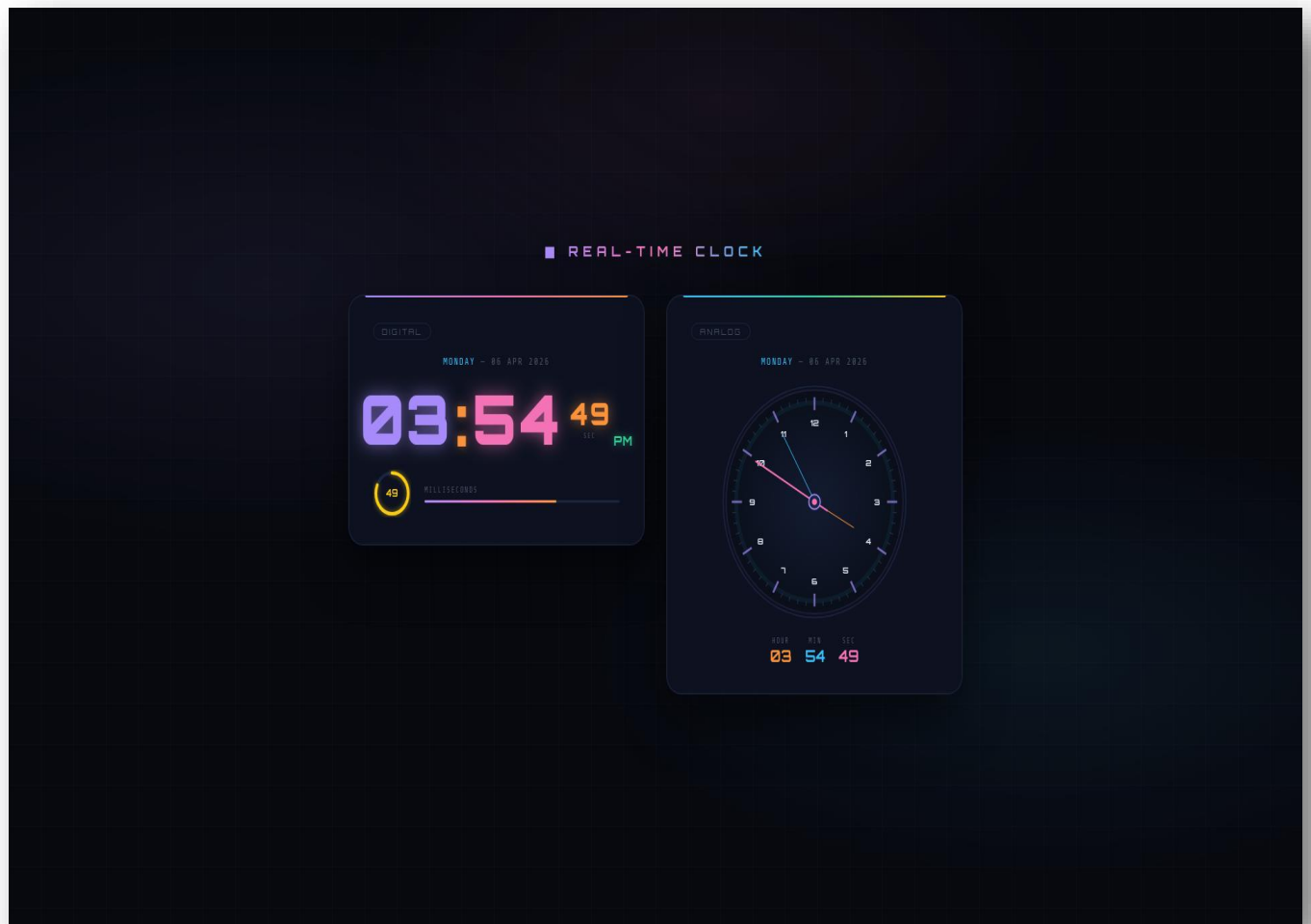
SUMMARY

DATE:- 06-04-2004

SRN:- PES1PG25CA147

NAME:- PAVAN G M

OUTPUT



CODE

Main.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"/>
<meta name="viewport" content="width=device-width,initial-scale=1"/>
<title>Real-Time Clock</title>
<link rel="stylesheet" href="clock.css"/>
</head>
<body>

<p class="page-title">&#9632; Real-Time Clock</p>
```

```

<div class="clocks-row">

<!-- === DIGITAL === -->
<div class="clock-card card-digital">
<span class="card-label">Digital</span>
<div class="date-row" id="d-date">—</div>

<div class="digital-display">
<div class="digit-group dg-h">
<span class="digit" id="dh1">0</span>
<span class="digit" id="dh2">0</span>
</div>
<span class="colon">:</span>
<div class="digit-group dg-m">
<span class="digit" id="dm1">0</span>
<span class="digit" id="dm2">0</span>
</div>
<div class="secs-block">
<div class="digit-group">
<span class="digit" id="ds1">0</span>
<span class="digit" id="ds2">0</span>
</div>
<span class="lbl">SEC</span>
</div>
<span class="ampm" id="dampm">AM</span>
</div>

<div class="dig-bottom">
<div class="prog-ring-wrap">
<canvas id="ring-canvas" width="56" height="56"></canvas>
<span class="prog-center" id="ring-num">00</span>
</div>
<div class="dig-bottom-right">
<div class="ms-label">MILLISECONDS</div>
<div class="ms-bar">
<div class="ms-fill" id="ms-fill" style="width:0%"></div>

```

```

</div>
</div>
</div>
</div>

<!-- ===== ANALOG ===== -->
<div class="clock-card card-analog">
<span class="card-label">Analog</span>
<div class="date-row" id="a-date">—</div>

<div class="analog-wrap">
<svg id="analog-svg" viewBox="0 0 260 260" xmlns="http://www.w3.org/2000/svg">
<defs>
<radialGradient id="face-grad" cx="50%" cy="50%" r="50%">
<stop offset="0%" stop-color="#141c32"/>
<stop offset="100%" stop-color="#0a0f1a"/>
</radialGradient>
<filter id="glow-v" x="-50%" y="-50%" width="200%" height="200%">
<feGaussianBlur stdDeviation="3" result="blur"/>
<feMerge><feMergeNode in="blur"/><feMergeNode in="SourceGraphic"/></feMerge>
</filter>
<filter id="glow-b" x="-50%" y="-50%" width="200%" height="200%">
<feGaussianBlur stdDeviation="3" result="blur"/>
<feMerge><feMergeNode in="blur"/><feMergeNode in="SourceGraphic"/></feMerge>
</filter>
</defs>

<!-- decorative rings -->
<circle cx="130" cy="130" r="126" fill="none" stroke="#1a2540" stroke-width="1.5"/>
<circle cx="130" cy="130" r="122" fill="none" stroke="rgba(167,139,250,.15)" stroke-width="1"/>
<!-- face -->
<circle cx="130" cy="130" r="120" fill="url(#face-grad)"/>
<!-- inner glow ring -->
<circle cx="130" cy="130" r="108" fill="none" stroke="rgba(56,189,248,.08)" stroke-width="5"/>

<!-- built by JS -->

```

```

<g id="a-ticks"></g>
<g id="a-nums"></g>

<!-- hands -->
<line id="hand-hour" class="hand hand-hour"
x1="130" y1="130" x2="130" y2="72" filter="url(#glow-v)"/>
<line id="hand-minute" class="hand hand-minute"
x1="130" y1="130" x2="130" y2="46" filter="url(#glow-b)"/>
<line id="hand-second" class="hand hand-second"
x1="130" y1="150" x2="130" y2="38"/>

<!-- center hub -->
<circle cx="130" cy="130" r="8" fill="#1a2540" stroke="#a78bfa" stroke-width="1.5"/>
<circle cx="130" cy="130" r="3.5" fill="#f472b6"/>
</svg>
</div>

<div class="analog-readout">
<div class="readout-item">
<div class="readout-label">HOUR</div>
<div class="readout-val rv-h" id="a-h">--</div>
</div>
<div class="readout-item">
<div class="readout-label">MIN</div>
<div class="readout-val rv-m" id="a-m">--</div>
</div>
<div class="readout-item">
<div class="readout-label">SEC</div>
<div class="readout-val rv-s" id="a-s">--</div>
</div>
</div>
</div>
</div>

</div><!-- .clocks-row -->

<script src="clock.js"></script>

```

```
</body>
</html>
```

Clock.css

```
@import
url('https://fonts.googleapis.com/css2?family=Orbitron:wght@400;700;900&family=Share+Tech+Mono&display=swap');

*,*::before,*::after{box-sizing:border-box;margin:0;padding:0}

:root{
  --bg: #07090f;
  --panel: #0e1220;
  --border: #1a2540;
  --c1: #a78bfa;
  --c2: #f472b6;
  --c3: #38bdf8;
  --c4: #fb923c;
  --c5: #34d399;
  --c6: #facc15;
  --text: #e2e8f0;
  --muted: #475569;
  --r: 22px;
}

body{
  background:var(--bg);
  color:var(--text);
  font-family:'Share Tech Mono',monospace;
  min-height:100vh;
  display:flex;
  flex-direction:column;
  align-items:center;
  justify-content:center;
```

```
gap:40px;
padding:40px 20px;
overflow-x:hidden;
}

body::before{
content:";
position:fixed;inset:0;
background:
radial-gradient(ellipse 60% 40% at 20% 30%,rgba(167,139,250,.07) 0%,transparent 70%),
radial-gradient(ellipse 50% 35% at 80% 70%,rgba(56,189,248,.07) 0%,transparent 70%),
radial-gradient(ellipse 40% 30% at 55% 10%,rgba(244,114,182,.06) 0%,transparent 70%);
pointer-events:none;z-index:0;
}

body::after{
content:";
position:fixed;inset:0;
background-image:
linear-gradient(rgba(167,139,250,.025) 1px,transparent 1px),
linear-gradient(90deg,rgba(167,139,250,.025) 1px,transparent 1px);
background-size:48px 48px;
pointer-events:none;z-index:0;
}

/* — Page title — */
.page-title{
font-family:'Orbitron',sans-serif;
font-size:clamp(.8rem,2.5vw,1.1rem);
font-weight:700;
letter-spacing:.4em;
text-transform:uppercase;
background:linear-gradient(90deg,var(--c1),var(--c2),var(--c3));
-webkit-background-clip:text;
-webkit-text-fill-color:transparent;
background-clip:text;
```

```
z-index:1;
}

/* — Clocks row — */
.clocks-row{
display:flex;
flex-wrap:wrap;
gap:32px;
justify-content:center;
align-items:flex-start;
width:100%;
max-width:960px;
z-index:1;
}

/* — Shared card — */
.clock-card{
background:var(--panel);
border:1px solid var(--border);
border-radius:var(--r);
padding:32px 36px;
display:flex;
flex-direction:column;
align-items:center;
gap:18px;
position:relative;
flex:1 1 360px;
max-width:440px;
box-shadow:0 0 0 1px rgba(255,255,255,.03) inset,0 24px 60px rgba(0,0,0,.5);
}

.clock-card::before{
content:"";
position:absolute;top:0;left:24px;right:24px;height:2px;
border-radius:0 0 4px 4px;
}
```

```

.card-digital::before{background:linear-gradient(90deg,var(--c1),var(--c2),var(--c4));}
.card-analog::before {background:linear-gradient(90deg,var(--c3),var(--c5),var(--c6));}

.card-label{
font-family:'Orbitron',sans-serif;
font-size:.6rem;
letter-spacing:.25em;
padding:3px 12px;
border-radius:99px;
border:1px solid var(--border);
color:var(--muted);
text-transform:uppercase;
align-self:flex-start;
}

/* — Date row — */
.date-row{
font-size:.75rem;
letter-spacing:.12em;
color:var(--muted);
text-transform:uppercase;
}
.date-row .hi{color:var(--c3);}

/* ===== DIGITAL ===== */
.digital-display{
display:flex;
align-items:center;
gap:6px;
line-height:1;
}
.digit-group{display:flex;gap:1px;}

.digit{
font-family:'Orbitron',sans-serif;
font-size:clamp(2.8rem,10vw,5rem);

```

```
font-weight:900;
width:1ch;
text-align:center;
}
.dg-h .digit{color:var(--c1);text-shadow:0 0 20px rgba(167,139,250,.6);}
.dg-m .digit{color:var(--c2);text-shadow:0 0 20px rgba(244,114,182,.6);}

.colon{
font-family:'Orbitron',sans-serif;
font-size:clamp(2.8rem,10vw,5rem);
font-weight:900;
color:var(--border);
line-height:1;
align-self:flex-start;
margin-top:4px;
animation:blink 1s step-start infinite;
}
@keyframes blink{50%{color:var(--c4);text-shadow:0 0 12px rgba(251,146,60,.5);}}

.secs-block{
display:flex;flex-direction:column;
align-items:center;gap:4px;
margin-left:4px;
}
.secs-block .digit{
font-size:clamp(1.4rem,4.5vw,2.2rem);
color:var(--c4);
text-shadow:0 0 16px rgba(251,146,60,.6);
}
.secs-block .lbl{font-size:.5rem;letter-spacing:.2em;color:var(--muted);}

.ampm{
font-family:'Orbitron',sans-serif;
font-size:clamp(.75rem,2vw,.95rem);
font-weight:700;
color:var(--c5);
```

```

letter-spacing:.08em;
align-self:flex-end;
margin-bottom:8px;
text-shadow:0 0 10px rgba(52,211,153,.5);
}

.prog-ring-wrap{position:relative;width:56px;height:56px;}
.prog-ring-wrap canvas{position:absolute;inset:0;}
.prog-center{
position:absolute;inset:0;
display:flex;align-items:center;justify-content:center;
font-family:'Orbitron',sans-serif;
font-size:.7rem;font-weight:700;
color:var(--c6);
}

.ms-label{font-size:.6rem;letter-spacing:.15em;color:var(--muted);margin-bottom:6px;}
.ms-bar{width:100%;height:3px;background:var(--border);border-radius:99px;overflow:hidden;}
.ms-fill{height:100%;background:linear-gradient(90deg,var(--c1),var(--c2),var(--c4));border-radius:99px;will-change:width;}

.dig-bottom{display:flex;align-items:center;gap:20px;width:100%;}
.dig-bottom-right{flex:1;}

/*=====ANALOG=====*/
.analog-wrap{
position:relative;
width:clamp(220px,55vw,280px);
height:clamp(220px,55vw,280px);
}
#analog-svg{width:100%;height:100%;overflow:visible;}

.hand{stroke-linecap:round;}
.hand-hour {stroke:var(--c4);stroke-width:6;opacity:.9;}
.hand-minute{stroke:var(--c3);stroke-width:4;opacity:.9;}
.hand-second{stroke:var(--c2);stroke-width:2;}

```

```

.tick-maj{stroke:var(--c1);stroke-width:2.5;opacity:.7;}
.tick-min{stroke:var(--muted);stroke-width:1.2;opacity:.4;}
.hour-lbl{
font-family:'Orbitron',sans-serif;
font-size:10px;font-weight:700;
fill:var(--text);
text-anchor:middle;dominant-baseline:center;
}

.analog-readout{display:flex;gap:20px;}
.readout-item{text-align:center;}
.readout-label{font-size:.55rem;letter-spacing:.2em;color:var(--muted);}
.readout-val{
font-family:'Orbitron',sans-serif;
font-size:1.2rem;font-weight:700;
}
.rv-h{color:var(--c4);}
.rv-m{color:var(--c3);}
.rv-s{color:var(--c2);}

```

Clock.js

```

/* — Build analog face — */
(function () {
const CX = 130, CY = 130, R = 114;
const ns = 'http://www.w3.org/2000/svg';
const tg = document.getElementById('a-ticks');
const ng = document.getElementById('a-nums');
const labels = [12, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11];

for (let i = 0; i < 60; i++) {
const ang = (i / 60) * 2 * Math.PI - Math.PI / 2;
const isMaj = i % 5 === 0;
const len = isMaj ? 14 : 5;

```

```

const x1 = CX + R * Math.cos(ang), y1 = CY + R * Math.sin(ang);
const x2 = CX + (R - len) * Math.cos(ang), y2 = CY + (R - len) * Math.sin(ang);

const line = document.createElementNS(ns, 'line');
line.setAttribute('x1', x1.toFixed(2)); line.setAttribute('y1', y1.toFixed(2));
line.setAttribute('x2', x2.toFixed(2)); line.setAttribute('y2', y2.toFixed(2));
line.setAttribute('class', isMaj ? 'tick-maj' : 'tick-min');
tg.appendChild(line);

if (isMaj) {
const rN = R - 28;
const txt = document.createElementNS(ns, 'text');
txt.setAttribute('x', (CX + rN * Math.cos(ang)).toFixed(2));
txt.setAttribute('y', (CY + rN * Math.sin(ang)).toFixed(2));
txt.setAttribute('class', 'hour-lbl');
txt.textContent = labels[i / 5];
ng.appendChild(txt);
}
}
})();

/* — Ring canvas — */
const ring = document.getElementById('ring-canvas');
const rctx = ring.getContext('2d');

function drawRing(sec) {
const cx = 28, cy = 28, r = 24;
rctx.clearRect(0, 0, 56, 56);
rctx.beginPath();
rctx.arc(cx, cy, r, 0, 2 * Math.PI);
rctx.strokeStyle = '#1a2540';
rctx.lineWidth = 4;
rctx.stroke();
if (sec > 0) {
rctx.beginPath();
rctx.arc(cx, cy, r, -Math.PI / 2, (sec / 60) * 2 * Math.PI - Math.PI / 2);

```

```

rctx.strokeStyle = '#facc15';
rctx.lineWidth = 4;
rctx.lineCap = 'round';
rctx.shadowColor = 'rgba(250,204,21,.6)';
rctx.shadowBlur = 6;
rctx.stroke();
rctx.shadowBlur = 0;
}
}

/* — Helpers — */
const pad = n => String(n).padStart(2, '0');
const DAYS = ['Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday'];
const MONTHS = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'];

/* — Main tick — */
function tick() {
const now = new Date();
const h24 = now.getHours();
const min = now.getMinutes();
const sec = now.getSeconds();
const ms = now.getMilliseconds();
const h12 = h24 % 12 || 12;

const dateHTML =
`<span class="hi">${DAYS[now.getDay()]}</span> &mdash; ` +
`${pad(now.getDate())} ${MONTHS[now.getMonth()]} ${now.getFullYear()}`;

/* DIGITAL */
const hStr = pad(h12), mStr = pad(min), sStr = pad(sec);
document.getElementById('dh1').textContent = hStr[0];
document.getElementById('dh2').textContent = hStr[1];
document.getElementById('dm1').textContent = mStr[0];
document.getElementById('dm2').textContent = mStr[1];
document.getElementById('ds1').textContent = sStr[0];
document.getElementById('ds2').textContent = sStr[1];

```

```
document.getElementById('dampm').textContent = h24 < 12 ? 'AM' : 'PM';
document.getElementById('d-date').innerHTML = dateHTML;
drawRing(sec);
document.getElementById('ring-num').textContent = pad(sec);
document.getElementById('ms-fill').style.width = (ms / 10).toFixed(1) + '%';

/* ANALOG */
const sDeg = (sec + ms / 1000) * 6;
const mDeg = (min + sec / 60) * 6;
const hDeg = (h24 % 12 + min / 60) * 30;
document.getElementById('hand-hour').setAttribute('transform', `rotate(${hDeg.toFixed(2)} 130 130)`);
document.getElementById('hand-minute').setAttribute('transform', `rotate(${mDeg.toFixed(2)} 130 130)`);
document.getElementById('hand-second').setAttribute('transform', `rotate(${sDeg.toFixed(3)} 130 130)`);
document.getElementById('a-date').innerHTML = dateHTML;
document.getElementById('a-h').textContent = pad(h12);
document.getElementById('a-m').textContent = mStr;
document.getElementById('a-s').textContent = sStr;
}

tick();
setInterval(tick, 40);
```